



PROFESSIONAL CERTIFICATION

Autodesk Inventor Training

The Autodesk Inventor Course teaches parametric 3D design for mechanical parts. Learners explore modeling assemblies and drawings. Practical training includes simulation sheet metal and plastic parts design for product engineering and manufacturing processes.



35-43

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.

Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★ Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

Autodesk Inventor Course

It is a way to learn mechanical design with Inventor tool.

Course Description

This course provides Inventor training for mechanical designers. Master iParts iAssemblies and content center usage. Projects focus on machine design tooling and automation for efficient product development in manufacturing industries.

Duration

35-43

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Master Inventor interface.
- ✓ Create parametric parts.
- ✓ Build dynamic assemblies.
- ✓ Perform stress analysis.
- ✓ Design sheet metal.
- ✓ Generate detailed drawings.
- ✓ Enhance design automation.

Target Audience

- Mechanical Engineers design.
- Fabrication pros modeling.
- Automobile in manufacturing.

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic mechanical knowledge required.
- Computer skills needed.
- Interest in parametric CAD.

Autodesk Inventor Course

Begin with Inventor basics and part creation.

Intermediate

- › Covers all Intermediate Levels
- › Precise Geometry Creation
- › Basic Parameters & Standards
- › Understanding of Tools & Uses
- › Intermediate Project Standard
- › 1 Project at Completion

Duration
35 Hrs

Professional

- › Covers all Professional Modules
- › 16 Standard Professional Modules
- › Learn with Industry Standard
- › Professional Tools and Uses
- › Multiple Professional Projects
- › 3 Projects at Completion

Duration
39 Hrs

Advanced (Customized)

- › Covers All Professional Modules
- › 16+ Advance Professional Module
- › Co-Ordination and Adv. Parameters
- › Shop Drawings & As Built Drawings
- › Govt. Standard and Submission
- › 4 Projects in Different Categories

Duration
43 Hrs

Autodesk Inventor Course

Begin with Inventor basics and part creation.

01 Navigating the Autodesk Inventor User Interface

- > Explore ribbon tabs panels and browser structure
- > Customize Quick Access Toolbar and keyboard shortcuts
- > Manage projects and file referencing workflows
- > Utilize view navigation and display styles
- > Access iProperties help system and learning resources

02 Creating and Constraining 2D Sketches

- > Draw precise geometry using lines arcs and splines
- > Apply geometric and dimensional constraints systematically
- > Use sketch patterns mirrors and projections
- > Edit and repair under-constrained or over-constrained sketches
- > Incorporate construction geometry and reference planes

03 Building Parametric Solid Parts

- > Extrude revolve sweep and loft profiles into features
- > Apply fillets chamfers holes and thread features
- > Create work features planes axes and points
- > Use direct editing tools for imported geometry
- > Inspect part mass properties and section views

04 Working with Placed and Derived Features

- > Insert patterned features rectangular circular and mirror
- > Create rib draft shell and split features
- > Derive parts from existing models or assemblies
- > Use multi-body solids and combine operations
- > Manage feature dependencies and edit history

05 Designing Sheet Metal Components

- > Define sheet metal defaults and unfold rules
- > Create base flanges contour flanges and hems
- > Add bends corner seams and relief options
- > Generate flat patterns and bend tables
- > Convert solid models to sheet metal parts

06 Creating and Managing Assemblies

- > Place and constrain components using mates and constraints
- > Define grounded flexible and adaptive parts
- > Insert subassemblies and drive constraints
- > Perform interference detection and contact analysis
- > Manage assembly relationships in the browser

Autodesk Inventor Course

Begin with Inventor basics and part creation.

07 Working with Assembly Joints and Motion

- > Apply rigid revolute slider and cylindrical joints
- > Insert limits angular and positional constraints
- > Drive joint motion and create positional representations
- > Use contact sets for realistic interaction
- > Generate exploded views and presentation files

08 Creating Frame Generator Structures

- > Populate frames from sketches or existing geometry
- > Insert standard frame members and profiles
- > Apply miter notch and trim to frame ends
- > Generate cut lists and material takeoffs
- > Edit frame members and rebuild assemblies

09 Performing Basic Stress Analysis

- > Assign materials and apply fixed and frictionless constraints
- > Define structural loads forces moments pressures
- > Run static stress simulations and interpret results
- > Generate safety factor and displacement plots
- > Refine mesh and convergence settings

10 Preparing Models for Manufacturing CAM Basics

- > Create manufacturing environments and tool libraries
- > Define setups for 2.5-axis milling operations
- > Generate facing contouring and pocket toolpaths
- > Simulate toolpaths and material removal
- > Post-process NC code for CNC machines

11 Creating Detailed Part and Assembly Drawings

- > Generate base projected section and detail views
- > Place automated centerlines dimensions and tolerances
- > Add hole tables thread notes and surface finish symbols
- > Create bill of materials and parts lists
- > Customize drawing templates and title blocks

12 Implementing iParts and iAssemblies

- > Create table-driven iParts with varying parameters
- > Define member files and custom iProperties
- > Build configurable iAssemblies with table control
- > Place iLogic rules for basic automation
- > Manage and suppress features in configurations

Autodesk Inventor Course

Begin with Inventor basics and part creation.

13 Using iLogic for Design Automation

- › Create parameters and rules using iLogic editor
- › Drive dimensions suppress features and trigger events
- › Embed external rules and global forms
- › Automate drawing creation and updates
- › Debug and test iLogic code effectively

14 Conducting Generative Design Exploration

- › Set up generative design studies with preserve/obstacle regions
- › Define manufacturing methods additive subtractive casting
- › Specify design objectives minimize mass maximize stiffness
- › Generate and evaluate multiple outcome sets
- › Convert promising outcomes to parametric Inventor models

15 Optimizing Designs for Additive Manufacturing

- › Prepare models for 3D printing using additive setup tools
- › Orient parts and generate intelligent support structures
- › Analyze overhangs bridging and build orientation
- › Incorporate lattice and lightweighting features
- › Export STL OBJ or 3MF files with build preparation

16 Producing Complete Manufacturing Documentation

- › Create production-ready drawings with GD&T annotations
- › Generate exploded assembly views and animation sequences
- › Produce weldment symbols and shop drawings
- › Export STEP IGES SAT and PDF files for collaboration
- › Package design data for manufacturing handoff

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

Early Bird Discount

Free

Demo Session



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